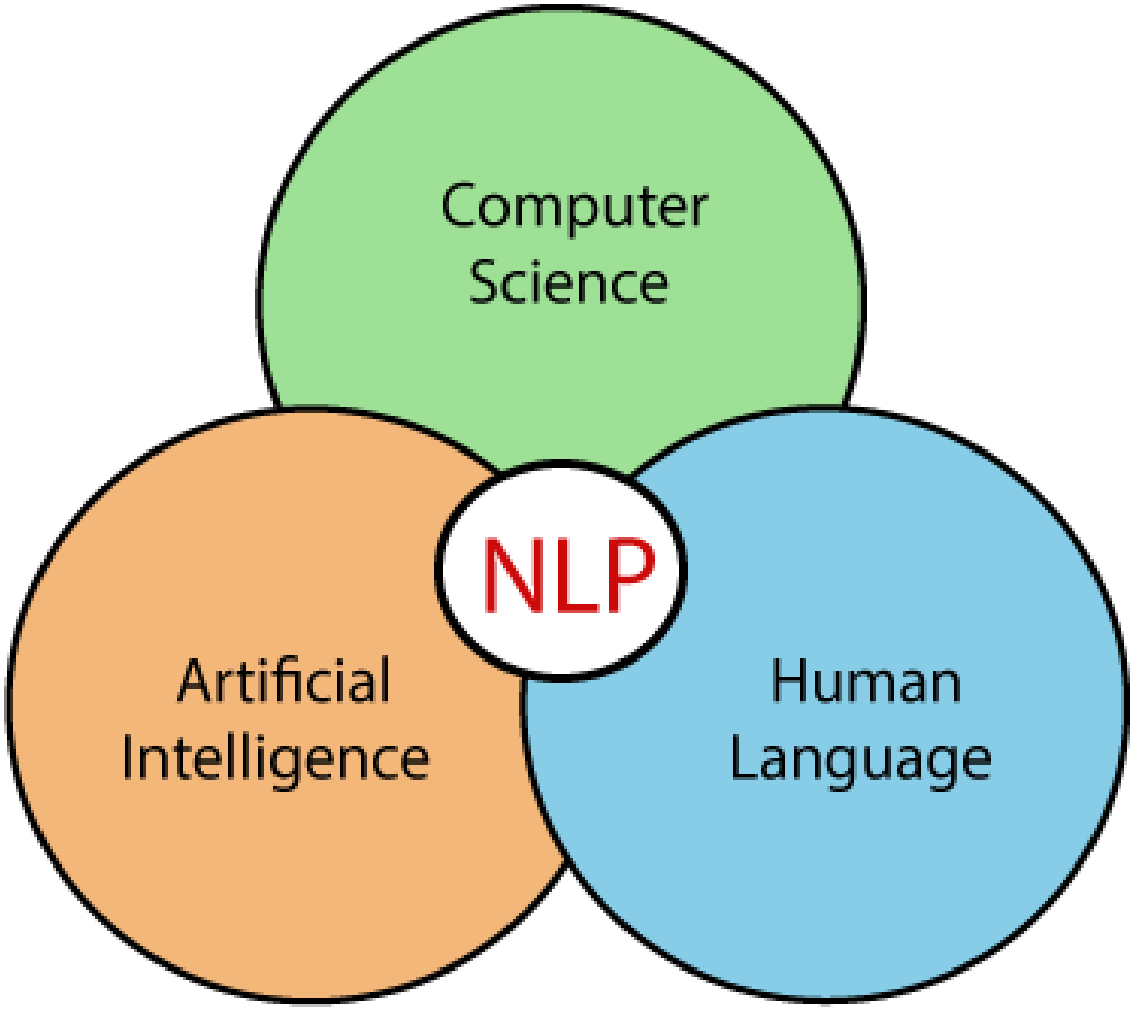
Module 6

Applications of AI and Current State of research in AI	
6.1	**Natural Language Processing:** Steps in NLP, Parsing and semantic interpretation, discourse and pragmatic processing, implementation aspects of syntactic analysis
6.2	**Expert system:** components of expert system, Expert Systems vs Traditional System. Characteristics of expert systems, roles in ES implementation, ES implementation process, applications, advantages and limitations of ES
6.3	Case studies based on current research in AI **Prompt engineering:** types of prompts, Prompt Failures & Biases, applications **AI copilot:** GitHub copilot **Retrieval Augmented Generation**[RAG], Real world examples of RAG


Computer
Science
NLP
Artificial
Intelligence
Human
Language

Natural Language Processing (NLP)

- NLP refers to AI method of communicating with an intelligent systems using a natural language such as English.
- Used in →Intelligent system like robot to perform as per your instructions, when you want to hear decision from a dialogue based clinical expert system, etc.
- Involves making computers to perform useful tasks with the natural languages humans use.
- The input and output of an NLP system can be −
  - Speech
  - Written Text

Components of NLP

1. NATURAL LANGUAGE UNDERSTANDING (NLU)

- Helps the machine to understand and analyse human language by extracting the **metadata from content such as concepts, entities, keywords**, emotion, relations, and semantic roles.
  - Used in Business applications to understand the customer's problem in both spoken and written language.
- NLU involves the following tasks -
  - Maps the given input into useful representation.
  - Analyzes different aspects of the language.

2. NATURAL LANGUAGE GENERATION (NLG)

- Natural Language Generation (NLG) acts as a **translator** that **converts the computerized data into natural language** representation.
- It mainly involves Text planning, Sentence planning, and Text Realization.

Steps to build an NLP

- Step1: **Sentence Segmentation-** Paragraph→ Sentences
- Step2: **Word Tokenization- Sentences** → Words/Tokens
- Step3: **Stemming-**Normalize words[intelligence, intelligent,& intelligently→root word "intelligen.“]
- Step 4: L**emmatization-**Grouping [Stemming]
- Step 5: **Identifying Stop Words-** removing is , and , a , the…
- Step 6: **Dependency Parsing**- relation of words in the sentence
- Step 7: P**OS** tags-Noun, verb, adverb, and Adjective
- Step 8: **Named Entity Recognition (NER)-**detecting the named entity such as person name, movie name, organization name, or location.; **Steve Jobs** introduced I phones
- Step 9: **Chunking-**collect the individual piece of information and grouping them into bigger pieces of sentences.

Applications of NLP

Question Answering

- Question Answering focuses on building systems that automatically answer the questions asked by humans in a natural language.

Spam Detection

- Spam detection is used to detect unwanted e-mails getting to a user's inbox.

Sentiment Analysis

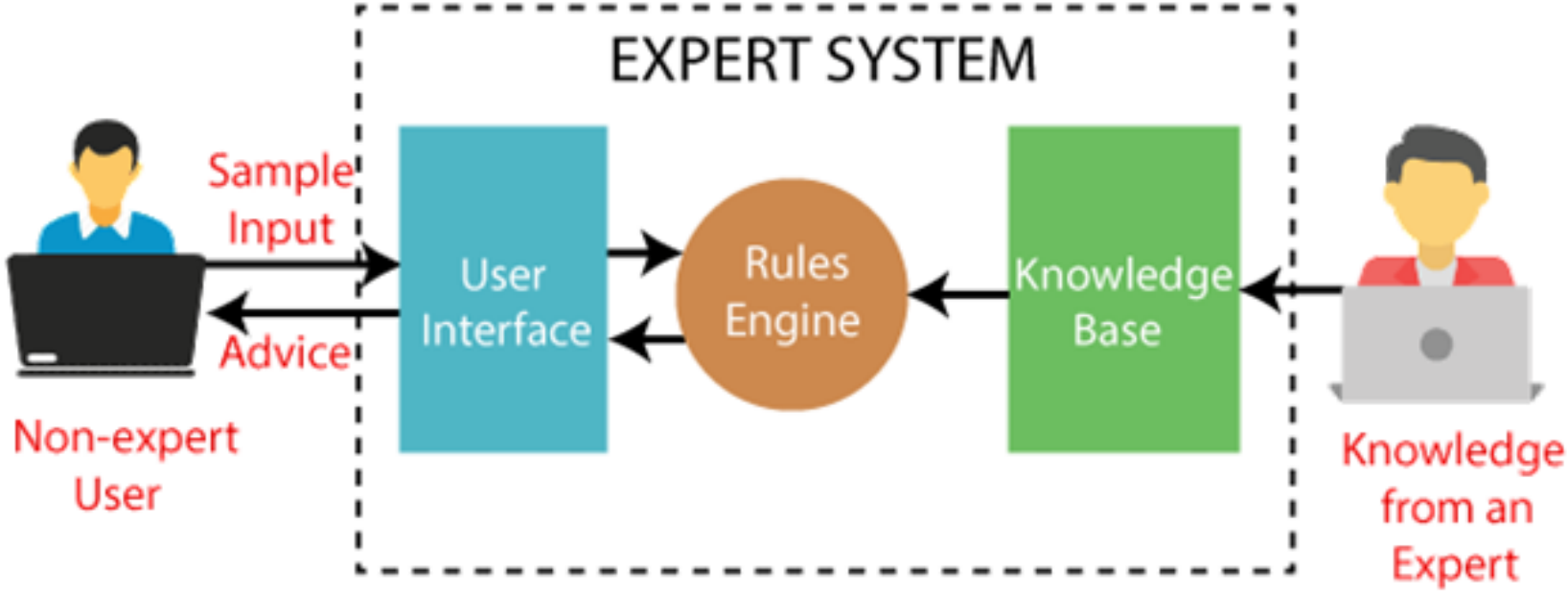
- Sentiment Analysis is also known as **opinion mining**
  - Analyse the attitude, behaviour, and emotional state of the sender.
- **Google Translator, Spelling correction, Speech Recognition , ChatBot,etc…..**

Expert systems (ES)

- One of the prominent research domains of AI.
- It is introduced by the researchers at Stanford University, Computer Science Department.

What are Expert Systems?

- The expert systems are the computer applications developed to solve complex problems in a particular domain, at the level of extra-ordinary human intelligence and expertise.


EXPERT SYSTEM
Sample Input
Advice
Non-expert User
User Interface
Rules Engine
Knowledge Base
Knowledge from an Expert

Characteristics of Expert System

- **High Performance:** The expert system provides high performance for solving any type of complex problem of a specific domain with high efficiency and accuracy.
- **Understandable:** It responds in a way that can be easily understandable by the user. It can take input in human language and provides the output in the same way.
- **Reliable:** It is much reliable for generating an efficient and accurate output.
- **Highly responsive:** ES provides the result for any complex query within a very short period of time.

Components of Expert System

- An expert system mainly consists of three components:
  - **User Interface**
  - **Inference Engine-Rules(Fwd chaining&Bckward chaining)**
  - **Knowledge Base**

Benefits of Expert Systems

- **Availability** − They are easily available due to mass production of software.
- **Less Production Cost** − Production cost is reasonable. This makes them affordable.
- **Speed** − They offer great speed. They reduce the amount of work an individual puts in.
- **Less Error Rate** − Error rate is low as compared to human errors.
- **Reducing Risk** − They can work in the environment dangerous to humans.
- **Steady response** − They work steadily without getting motional, tensed or fatigued.

Expert Systems Limitations

- No technology can offer easy and complete solution.
- Large systems are costly, require significant development time, and computer resources.
- ESs have their limitations which include −
  - Limitations of the technology
  - Difficult knowledge acquisition
  - ES are difficult to maintain
  - High development costs

Case studies based on current research in AI

- **Prompt engineering:** types of prompts, Prompt Failures & Biases, applications
- **AI copilot:** GitHub copilot
- **Retrieval Augmented Generation**[RAG], Real world examples of RAG

Case Studies in AI (Current Research)

- Healthcare: Disease prediction
- Finance: Fraud detection
- Autonomous vehicles
- AI chatbots and assistants

Case Study Example

AI in Healthcare:

- Detects cancer from images
- Uses deep learning models
- Improves diagnosis accuracy

Prompt Engineering

- Designing effective inputs for AI models
- Improves output quality
- Used in ChatGPT, Copilot

Types of Prompts

- Zero-shot
- Few-shot
- Chain-of-thought
- Instruction-based prompts

Prompt Failures

- Ambiguous prompts
- Hallucinations
- Incomplete outputs
- Misinterpretation

Bias in Prompts

- Training data bias
- Cultural bias
- Gender bias
- Can lead to unfair results

Applications of Prompt Engineering

- Content generation
- Code generation
- Customer support
- Automation tasks

AI Copilot - GitHub Copilot

- AI coding assistant
- Suggests code in real-time
- Works in VS Code, JetBrains

Features of GitHub Copilot

- Auto-completion
- Code suggestions
- Supports multiple languages
- Improves productivity

Retrieval Augmented Generation (RAG)

- Combines LLM with external data
- Fetches relevant info before generating answer
- Improves accuracy

How RAG Works

- User query → Retrieve data → Generate answer
- Uses vector databases
- Enhances factual correctness

Real-world Examples of RAG

- Chatbots with PDFs
- Search engines
- Customer support systems
- Enterprise AI assistants

Conclusion

- AI is evolving rapidly
- Prompt engineering and RAG are key technologies
- Copilot boosts developer productivity